ETTJO: An Event-Triggered Two-Timescale Framework for Joint Caching, Trajectory, and Offloading in Maritime MEC

Kai Yin, Yuxiang Zhao, Ruiting Wang, Xinxin Ye, Ting Gan, and Zongpeng Li, *Senior Member, IEEE*

***Abstract*—In maritime mobile edge computing (MEC) networks, uncrewed aerial vehicles (UAVs) serve as aerial relays to overcome offshore coverage limitations. However, sustainable UAV operation is challenged by the tight coupling among trajectory planning, content caching, and task offloading under strict energy constraints and stochastic maritime channels. Conventional periodic optimization further incurs excessive backhaul overhead and computational costs. To address these issues, we propose ETTJO, an event-triggered two-timescale joint optimization framework. Leveraging Lyapunov optimization, ETTJO decouples long-term energy constraints from instantaneous decisions without future network information. It further decomposes the problem into two timescales: event-triggered caching updates content only when service popularity changes significantly, while a hybrid short-timescale control algorithm integrates multiagent deep reinforcement learning (DRL) for continuous trajectory control with matching theory for discrete task offloading. We provide a theoretical analysis to support the proposed framework. Extensive simulations demonstrate that ETTJO achieves lower latency and improved energy efficiency compared with representative baselines.**

***Index Terms*—Deep reinforcement learning (DRL), event-triggered caching, Lyapunov optimization, maritime mobile edge computing (MEC), task offloading.**

NOMENCLATURE

$\mathcal{U}, \mathcal{M}, \mathcal{S}$	Sets of UAVs, MIoTs, and services.
u, m, s, t	Indices of UAV, MIoT, service, and slot.
$U+1$	Index of the ground station (GS).
C_u	Cache capacity of UAV u.
f_u, f_g	CPU frequencies of UAV u and GS g.
$c_s, n^t_{m,s}, r_s$	Service size, task size, and computation intensity.
$T, \Delta t$	Number and duration of short-timescale slots.
$d_{\min}$	Minimum safe distance between UAVs.
ϵ, χ	Link reliability thresholds.
$\mathbf{q}^t_u, \mathbf{q}^t_m, \mathbf{q}_g$	Coordinates of UAV u, MIoT m, and GS.
B_0, B_1	Bandwidths of MIoT–UAV and UAV–GS links.
$b^t_{u,s}$	Binary cache indicator for service s at UAV u.
$o^t_{m,s,u}$	Offloading indicator for task (m, s) to node u.
$k^t_{m,s,u}$	Binary relay indicator through UAV u to the GS.
$\mathbf{v}^t_u$	Velocity vector of UAV u.
v^t_u	Speed magnitude of UAV u, $v^t_u = \|\mathbf{v}^t_u\|_2$.
$R^t_{m,u}, R^t_{u,g}$	MIoT–UAV and UAV–GS transmission rates.
L^t_u, L^t_{sys}	UAV-associated latency and system latency.
E^t_u	Total energy consumption of UAV u.
$l^t_{m,u}, h^t_{m,s}$	Link reliability and cache-hit ratio.

I. INTRODUCTION

WITH the rapid advancement of Maritime IoT (MIoT), a gap has emerged between the surging data demands of heterogeneous devices—ranging from environmental monitoring buoys to autonomous vessels—and the limited transmission capacity of offshore networks [1]. Unlike terrestrial networks, maritime environments suffer from extreme isolation, characterized by sparse infrastructure coverage and extended distances to shore. These adverse conditions, compounded by deep fading and wave occlusion, frequently disrupt MIoT-to-shore communication, resulting in high latency and unreliable data delivery [2].

To address these coverage limitations, the integration of uncrewed aerial vehicles (UAVs) with mobile edge computing (MEC) has emerged as a flexible and cost-effective solution [3], [4]. Acting as aerial edge nodes, UAVs can provide flexible computation offloading and data relaying, significantly alleviating the processing burden on resource-constrained maritime devices [5]. While UAV-assisted MEC has achieved substantial success in terrestrial and vehicular networks [6], [7], [8], transplanting these frameworks to the maritime domain is nontrivial. Unlike grid-powered terrestrial infrastructures, maritime UAVs operate under stringent onboard energy budgets, where propulsion energy dominates consumption. Moreover, the high dynamism of maritime channels and vessel mobility

Received 21 May 2026; accepted 8 June 2026. Date of publication 11 June 2026; date of current version 7 September 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62472328 and in part by the Key Research and Development Program of Hubei Province under Grant 2023BAB170. *(Kai Yin and Yuxiang Zhao are co-first authors.) (Corresponding author: Ting Gan.)*

Kai Yin, Yuxiang Zhao, Ruiting Wang, and Ting Gan are with the School of Computer Science, Wuhan University, Wuhan 430072, China (e-mail: yinkai@whu.edu.cn; yuxiangzhao@whu.edu.cn; wangrt@whu.edu.cn; ganting@whu.edu.cn).

Xinxin Ye is with Nanjing Les Information Technology Company Ltd., Nanjing 210014, China (e-mail: ye_xx@les.cn).

Zongpeng Li is with the Institute for Network Sciences and Cyberspace, Tsinghua University, Beijing 100084, China (e-mail: zongpeng@tsinghua.edu.cn).

This article has supplementary downloadable material available at https://doi.org/10.1109/JIOT.2026.3702711, provided by the authors.

Digital Object Identifier 10.1109/JIOT.2026.3702711

2327-4662 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Indian Institute of Petroleum and Energy. Downloaded on October 26,2026 at 12:03:34 UTC from IEEE Xplore. Restrictions apply.

introduces severe uncertainty, undermining the performance of conventional deterministic optimization strategies [9].

Realizing an efficient UAV-assisted MEC system in such dynamic environments requires addressing three coupled challenges. 1) *Cost-Prohibitive Cache Updating:* Limited onboard storage restricts UAVs to caching only critical content, while stochastic MIoT mobility causes rapid service-demand fluctuations. Frequent remote updates incur prohibitive backhaul costs [10]. 2) *Energy-Constrained Trajectory Planning:* Aggressive maneuvering to maintain connectivity with dispersed vessels consumes excessive propulsion energy, critically limiting network lifetime [11]. 3) *Coupled Task Offloading:* Offloading decisions strictly depend on instantaneous cache availability and link quality [12]. These challenges are tightly interwoven: caching dictates offloading feasibility, trajectory determines link quality and energy consumption, and strict energy budgets constrain all three decisions.

To tackle these challenges, we propose the event-triggered two-timescale joint optimization (ETTJO) framework for maritime UAV-assisted MEC. Unlike periodic or single-timescale designs, ETTJO updates cache placement on a long timescale only when service demand changes significantly, while optimizing UAV trajectory and task offloading on a short timescale based on real-time network states. The main contributions are summarized as follows.

1) We formulate a joint caching, trajectory, and offloading problem for maritime UAV-assisted MEC, aiming to minimize long-term average system latency under cache-capacity, mobility, offloading feasibility, and long-term UAV energy constraints.
2) We develop a Lyapunov-guided two-timescale optimization framework. Lyapunov virtual energy queues transform long-term UAV energy constraints into online per-slot optimization problems, enabling a latency-energy trade-off without future information. Event-triggered caching reduces unnecessary cache updates, while short-timescale control adapts trajectory and offloading decisions.
3) We design a hybrid solution consisting of limited space greedy caching algorithm (LSGCA) for cache placement, HASAC for continuous multi-UAV trajectory control, and virtual port deferred acceptance (VPDA) for discrete task offloading.
4) We provide a theoretical analysis of the proposed framework and conduct simulations against representative baselines to evaluate system latency, energy consumption, cache performance, and operational endurance.

II. Related Work

A. Cache-Assisted MEC

Content caching represents an effective approach to alleviating backhaul pressure in MEC systems. Early works, such as the studies by Wu et al. [13] and Malik et al. [14], utilized submodular functions and greedy-like algorithms [15] based on marginal gains to optimize cache placement. However, the high mobility and stringent energy constraints of UAV-assisted networks render these approaches insufficient. To address UAV-specific dynamics, Wu et al. [16] employed multiagent deep reinforcement learning (DRL) for autonomous cache decisions, while Zhang et al. [17] investigated caching decisions in tandem with trajectory scheduling. Despite their flexibility, these DRL-based methods generally lack rigorous theoretical analysis regarding optimality bounds or convergence stability. More recently, Maale et al. [18] integrated federated learning with knowledge distillation to solve the joint optimization problem. While Zhang et al. [19] tailored a Top-K popularity strategy for maritime MEC, their approach mainly relies on offline popularity information and is less suitable for real-time stochastic maritime environments.

B. Task Offloading

In the realm of maritime and UAV-assisted MEC, Lu et al. [11] established a secure UAV-relay framework by jointly optimizing partial task offloading and trajectory to mitigate eavesdropping risks. To address task uncertainty and resource heterogeneity, You et al. [20] formulated offloading as a Markov game, employing DRL to minimize execution time. Recent DRL-based studies also investigate dynamic computation offloading in UAV-assisted MEC networks [21]. Similarly, Li et al. [22] proposed an integrated framework that maximizes data collection rates via joint trajectory and resource optimization. Furthermore, Wei et al. [5] prioritized energy efficiency by leveraging shipping lane information for long-term channel prediction to optimize power allocation. However, these studies primarily focus on offloading, trajectory, or resource allocation, while the influence of content caching on offloading feasibility and system latency is not fully considered.

C. Joint Service Caching and Task Offloading

Extensive literature investigates the joint optimization of content caching and task offloading in vehicular, aerial-ground, and edge-cloud MEC systems. Bai et al. [23] optimized task offloading and content caching in UAV-assisted Internet of Vehicles using DDPG, while Xie et al. [24] developed a DRL-based cache-assisted offloading strategy. Song et al. [25] jointly considered dependent task scheduling, resource allocation, and service caching in aerial-ground integrated MEC, while Li et al. [26] proposed two-timescale service caching and resource allocation with edge-cloud cooperation. These studies provide useful insights into caching-offloading coordination, but they do not fully address maritime UAV-assisted MEC, where vessel mobility, costly cache updating, trajectory-dependent link, and UAV energy constraints are tightly coupled. Addressing mobility, Qin et al. [27] studied coupled trajectory and caching optimization, yet their offline approach relies on future network states. Gao and Zhai [28] addressed mobility using Dinkelbach's method and successive convex approximation (SCA), but such optimization methods may incur high computational overhead in rapidly changing maritime environments. In contrast to these periodic or offline schemes, our work proposes an online event-triggered two-timescale framework that reduces unnecessary cache updates while adapting trajectory and offloading decisions to real-time maritime network states.

Authorized licensed use limited to: Indian Institute of Petroleum and Energy. Downloaded on October 26,2026 at 12:03:34 UTC from IEEE Xplore. Restrictions apply.

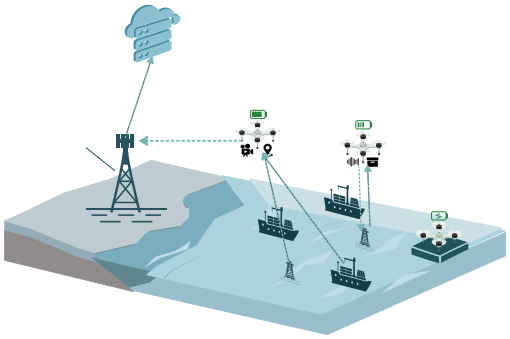


Fig. 1. Illustration of the maritime UAV-assisted MEC system.

III. System Overview

As illustrated in Fig. 1, we consider a maritime UAV-assisted MEC system deployed in coastal offshore areas. The system consists of U rotary-wing UAVs, denoted by $\mathcal{U}$, M mobile MIoT devices, denoted by $\mathcal{M}$, and a shore-side ground station (GS) g. MIoT devices, such as sensor buoys and autonomous vessels, generate service requests in areas with limited direct GS coverage. UAVs act as aerial edge servers, cache carriers, and relay nodes, while the GS provides sufficient computation capability for tasks that cannot be executed by UAVs.

The system operates over a finite horizon of T slots, each with duration Δt. At each slot t, MIoT devices move with velocity $\mathbf{v}_m^t$ and may request services in $\mathcal{S}$, while each UAV stores a subset of service functions subject to its cache capacity. If the requested service is cached by an accessible UAV with sufficient computation resources, the task is offloaded to the UAV for local execution; otherwise, it is relayed through a UAV to the GS for processing. Thus, the service process consists of UAV-assisted edge execution and UAV-to-GS relaying.

During each slot, the controller observes current requests, UAV/MIoT positions, cache states, and UAV energy, and assigns each task to either a cache-enabled UAV for execution or a selected UAV for relaying to the GS. Following slot-level scheduling models adopted in MEC systems [10], [29], [30], we focus on delay-sensitive and small-granularity MIoT tasks that can be completed within one slot when Δt is set to several or tens of seconds, such as status monitoring, alarm reporting, and environmental sensing. Delay-tolerant bulk-data services can be supported by extending the model with task queues and deadline-aware scheduling.

IV. System Model

This section establishes the system model for maritime UAV-assisted MEC. We model service caching, UAV mobility, communication, computation, and energy consumption, and then formulate the joint optimization problem for cache placement, trajectory control, and task offloading under UAV energy constraints.

A. Dynamic Caching

1) Cache Capacity Constraint: At slot t, $i_{m,s}^t \in \{0,1\}$ indicates whether MIoT device m requests service s. The task input size is $n_{m,s}^t$, while c_s denotes the storage size of service s. Each UAV $u \in [U]$ has a maximum capacity C_u. Let $b_{u,s}^t \in \{0,1\}$ indicate if service s (size c_s bits) is cached at time t. The storage must satisfy

$$\sum_{s=1}^{S} b_{u,s}^t c_s \le C_u \quad \forall u \in [U]. \tag{1}$$

2) Probabilistic Connectivity Model: Link reliability $l_{m,u}^t \in [0,1]$ between MIoT m and UAV u is a piecewise linear function of their distance $d_{m,u}^t = \|\mathbf{q}_u^t - \mathbf{q}_m^t\|_2$ relative to the coverage radius $d_u^{\max}$

$$l_{m,u}^t = \min\left(1, \frac{\max\left(0, d_u^{\max} - d_{m,u}^t\right)}{(1-\eta_{\text{link}})d_u^{\max}}\right) \tag{2}$$

where $\mathbf{q}_u^t$ and $\mathbf{q}_m^t$ are 3-D coordinates at time t. The threshold η_{link} ensures a stable link within $\eta_{\text{link}} d_u^{\max}$, beyond which connectivity decays linearly.

3) Cache Hit Ratio: A cache hit for service pair (m,s) occurs if the content is stored on at least one associated UAV. Neglecting computational limits, the hit probability $h_{m,s}^t$ is

$$h_{m,s}^t = 1 - \prod_{u=1}^{U}\left(1 - b_{u,s}^t l_{m,u}^t\right). \tag{3}$$

The system-wide weighted hit ratio at time t is normalized as

$$h^t = \frac{\sum_{m=1}^{M}\sum_{s=1}^{S} p_{m,s}^t h_{m,s}^t}{\sum_{m=1}^{M}\sum_{s=1}^{S} p_{m,s}^t} \tag{4}$$

where $p_{m,s}^t$ denotes the request probability for service s by user m.

4) Cache Optimization Problem: Given estimated request popularity and link reliabilities from (2), the triggered cache-placement problem is formulated as

$$\begin{aligned} (\mathbf{P1}) \max_{\{b_{u,s}^t\}} \quad & h^t \\ \text{s.t.} \quad & \sum_{s=1}^{S} b_{u,s}^t c_s \le C_u \quad \forall u \in [U] \\ & b_{u,s}^t \in \{0,1\} \quad \forall u \in [U] \quad \forall s \in [S]. \end{aligned} \tag{5}$$

Problem (**P1**) is an NP-hard 0/1 knapsack variant and serves as a long-timescale cache-placement subproblem rather than a complete control problem. In implementation, $b_{u,s}^t$ is updated only at triggered cache-update epochs and remains unchanged between two consecutive updates. Thus, (**P1**) provides cache-aware service availability for (**P2**), which further optimizes trajectory control, task offloading, and relay selection under the resulting cache state and long-term UAV energy constraints.

B. Communication, Mobility, and Energy Model

1) Velocity Constraint: Within each slot of duration Δt, UAVs and MIoT devices are assumed to move with constant velocities $\mathbf{v}_u^t$ and $\mathbf{v}_m^t$, respectively. The UAV position evolves as

$$\mathbf{q}_u^{t+1} = \mathbf{q}_u^t + \mathbf{v}_u^t \Delta t \quad \forall u \in [U] \quad \forall t. \tag{6}$$

The speed magnitude is defined as $v_u^t = \|\mathbf{v}_u^t\|_2$, and it satisfies

$$v_u^t \le V_u \quad \forall u \in [U] \quad \forall t. \tag{7}$$

Authorized licensed use limited to: Indian Institute of Petroleum and Energy. Downloaded on October 26,2026 at 12:03:34 UTC from IEEE Xplore. Restrictions apply.

2) Collision Avoidance: UAVs must maintain the minimum safe distance $d_{\min}$

$$\|\mathbf{q}_j^t - \mathbf{q}_k^t\|_2 \geq d_{\min} \quad \forall j,\ k \in [U],\ j \neq k \quad \forall t. \tag{8}$$

3) Link Constraint: To ensure that the joint disconnection probability of MIoT device m remains below ϵ, we impose

$$\prod_{u=1}^{U} \left(1 - l_{m,u}^t\right) \leq \epsilon \quad \forall m \quad \forall t \tag{9}$$

where $l_{m,u}^t$ is the soft link probability defined in (2), and ϵ limits the joint disconnection probability. The thresholds χ and ϵ are jointly set to ensure feasible serving links under normal operation. If no feasible physical service path exists, the task remains unmatched in VPDA and incurs a maximum-delay penalty.

4) UAV Propulsion Energy Model: For rotary-wing UAVs, propulsion dominates onboard energy. Following [31], [32], the steady-level flight power is modeled as:

$$P_{\text{fly},u}(v_u^t) = \underbrace{C_1 + C_2 (v_u^t)^2}_{\text{Blade Profile}} + \underbrace{C_3 \left(\sqrt{1 + \frac{(v_u^t)^4}{C_4^2}} - \frac{(v_u^t)^2}{C_4} \right)^{1/2}}_{\text{Induced}} + \underbrace{C_5 (v_u^t)^3}_{\text{Parasite}} \tag{10}$$

where $C_1, \ldots, C_5$ are UAV-dependent aerodynamic coefficients corresponding to blade profile, induced, and parasite power.

We adopt orthogonal multiple access, where B_0 and B_1 denote the allocated MIoT–UAV and UAV–GS bandwidths, respectively; co-channel interference is therefore omitted.

5) Sea-to-Air Channel: Due to maritime obstructions such as waves, link quality depends on the elevation angle $\theta_{m,u}^t$ [5], [20] calculated as $\theta_{m,u}^t = \arctan(|z_m^t - z_u^t| / d_{m,u}^t)$. Adopting the probabilistic LoS model [33], the path loss $\eta_{m,u}^t$ is

$$\eta_{m,u}^t = \frac{\eta_{\text{LOS}} - \eta_{\text{NLOS}}}{1 + a e^{-b(\theta_{m,u}^t - a)}} + 20 \log_{10}(d_{m,u}^t) + 20 \log_{10}\left(\frac{4\pi f_{u,m}}{300}\right) + \eta_{\text{NLOS}} \tag{11}$$

where a, b are environmental parameters, $f_{u,m}$ is the carrier frequency, and η_{LOS} and η_{NLOS} are LoS and NLoS attenuation factors, respectively. With $g_{m,u}^t = 10^{-\eta_{m,u}^t/10}$, the reliability-weighted uplink rate is

$$R_{m,u}^t = B_0\, l_{m,u}^t \log_2\left(1 + \frac{P_{\text{tx},m}^t g_{m,u}^t}{N_0}\right) \tag{12}$$

where $P_{\text{tx},m}^t$ denotes the transmission power and N_0 is the noise power, and $l_{m,u}^t$ weights the Shannon rate by link reliability.

6) Air-to-Shore Channel: Air-to-shore links are dominated by LoS propagation [34]. The channel gain follows the free-space path loss model $g_{u,g}^t = g_0 / \|\mathbf{q}_u^t - \mathbf{q}_g\|_2^2$, where g_0 is the reference gain at 1 m and $\mathbf{q}_g$ is the GS position. The transmission rate is

$$R_{u,g}^t = B_1 \log_2\left(1 + \frac{P_{\text{tx},u}^t g_{u,g}^t}{N_0}\right). \tag{13}$$

Downlink latency is neglected because computation results and control signaling are much smaller than uplink tasks.

C. Computation and Task Offloading Model

1) Decision Variables: At slot t, $o_{m,s,u}^t \in \{0,1\}$ denotes the offloading decision for task (m,s), where $u \in [U]$ represents a UAV and $u = U+1$ represents the GS. Each generated task is assigned to one processing path, and UAV-side execution is allowed only when the requested service is cached

$$o_{m,s,u}^t \leq b_{u,s}^t \quad \forall m \quad \forall s \quad \forall u \in [U] \quad \forall t$$
$$\sum_{u=1}^{U+1} o_{m,s,u}^t = i_{m,s}^t \quad \forall m \quad \forall s \quad \forall t. \tag{14}$$

Moreover, UAV-side offloading is feasible only when the corresponding link reliability satisfies the minimum requirement

$$\chi \cdot o_{m,s,u}^t \leq l_{m,u}^t \quad \forall m \quad \forall s, u \in [U] \quad \forall t. \tag{15}$$

The assignment constraints describe feasible physical service paths. If VPDA cannot match a generated task to any feasible path, the task is dropped and counted with a maximum-delay penalty $L_{\max} = \Delta t$.

2) Computation Model: The task (m,s) takes $r_s n_{m,s}^t$ CPU cycles, and the computation resource constraint of UAV u is

$$\sum_{m=1}^{M} \sum_{s=1}^{S} r_s n_{m,s}^t o_{m,s,u}^t \leq f_u \Delta t \quad \forall u \in [U] \quad \forall t. \tag{16}$$

The GS is assumed to have sufficient computation capability with $f_g \gg f_u$.

3) Latency Model: Let $k_{m,s,u}^t \in \{0,1\}$ indicate whether UAV u relays task (m,s) to the GS

$$\chi \cdot k_{m,s,u}^t \leq l_{m,u}^t \quad \forall m \quad \forall s, u \in [U] \quad \forall t \tag{17}$$
$$\sum_{u=1}^{U} k_{m,s,u}^t = o_{m,s,U+1}^t \quad \forall m \quad \forall s \quad \forall t. \tag{18}$$

The task latency is decomposed as $L_{m,s}^t = \sum_{u=1}^{U} L_{m,s,u}^t$, where for edge nodes $u \in [U]$

$$L_{m,s,u}^t = o_{m,s,u}^t n_{m,s}^t \left(\frac{1}{R_{m,u}^t} + \frac{r_s}{f_u}\right) + k_{m,s,u}^t n_{m,s}^t \left(\frac{1}{R_{m,u}^t} + \frac{1}{R_{u,g}^t} + \frac{r_s}{f_g}\right) \tag{19}$$

Let L_u^t denote the total latency associated with tasks computed by or relayed through UAV u

$$L_u^t = \sum_{m=1}^{M} \sum_{s=1}^{S} L_{m,s,u}^t. \tag{20}$$

The total system latency at time slot t is then defined as

$$L_{\text{sys}}^t = \sum_{u=1}^{U} L_u^t = \sum_{m=1}^{M} \sum_{s=1}^{S} \sum_{u=1}^{U} L_{m,s,u}^t. \tag{21}$$

4) Energy Consumption: The total energy expenditure of UAV u in slot t consists of computation, caching, flight, and communication components

$$E_{\text{cmp},u}^t = \sum_{m=1}^{M} \sum_{s=1}^{S} \kappa r_s n_{m,s}^t o_{m,s,u}^t$$

Authorized licensed use limited to: Indian Institute of Petroleum and Energy. Downloaded on October 26,2026 at 12:03:34 UTC from IEEE Xplore. Restrictions apply.

$$E^t_{\text{cache},u} = \sum_{s\in\Psi^t_u\setminus\Psi^{t-1}_u} \omega c_s$$
$$E^t_{\text{fly},u} = P_{\text{fly},u}\left(v^t_u\right)\Delta t$$
$$E^t_{\text{tx},u} = \sum_{m=1}^{M}\sum_{s=1}^{S} P^t_{\text{tx},u} k^t_{m,s,u} \frac{n^t_{m,s}}{R^t_{u,g}}. \tag{22}$$

Here, κ is the energy coefficient per CPU cycle, ω denotes the cache-update energy per bit, and Ψ^t_u is the service set cached at UAV u. The flight energy is determined by the propulsion power and slot duration. The communication energy only accounts for UAV-to-GS relay transmission, since MIoT-to-UAV transmission energy is consumed by MIoT devices and is not counted in UAV energy consumption.

Finally, the total energy consumption is

$$E^t_u = E^t_{\text{cmp},u} + E^t_{\text{cache},u} + E^t_{\text{fly},u} + E^t_{\text{tx},u}. \tag{23}$$

D. Joint Optimization Problem

Given the cache state determined by (**P1**), (**P2**) optimizes UAV trajectory control, task offloading, and relay selection under long-term UAV energy constraints

$$(\mathbf{P2}) \min_{\mathcal{G},\mathcal{O},\mathcal{K}} \quad \sum_{t=1}^{T} L^t_{\text{sys}}$$
$$\text{s.t. } \|\mathbf{v}^t_u\|_2 \le V_u \quad \forall u\in[U] \quad \forall t \tag{2a}$$
$$\|\mathbf{q}^t_j - \mathbf{q}^t_k\|_2 \ge d_{\min} \quad \forall j,\ k\in[U],\ j\ne k \quad \forall t \tag{2b}$$
$$\prod_{u=1}^{U}\left(1-l^t_{m,u}\right) \le \epsilon \quad \forall m \quad \forall t \tag{2c}$$
$$o^t_{m,s,u} \le b^t_{u,s} \quad \forall m \quad \forall s,\ u\in[U] \quad \forall t \tag{2d}$$
$$\chi\cdot o^t_{m,s,u} \le l^t_{m,u} \quad \forall m \quad \forall s,\ u\in[U] \quad \forall t \tag{2e}$$
$$\chi\cdot k^t_{m,s,u} \le l^t_{m,u} \quad \forall m \quad \forall s,\ u\in[U] \quad \forall t \tag{2f}$$
$$\sum_{u=1}^{U} k^t_{m,s,u} = o^t_{m,s,U+1} \quad \forall m \quad \forall s \quad \forall t \tag{2g}$$
$$\sum_{u=1}^{U+1} o^t_{m,s,u} = i^t_{m,s} \quad \forall m \quad \forall s \quad \forall t \tag{2h}$$
$$\sum_{m=1}^{M}\sum_{s=1}^{S} r_s n^t_{m,s} o^t_{m,s,u} \le f_u\Delta t \quad \forall u\in[U] \quad \forall t \tag{2i}$$
$$\sum_{t=1}^{T} E^t_u \le E^{\max}_u \quad \forall u\in[U] \tag{2j}$$
$$o^t_{m,s,u}\in\{0,1\} \quad \forall m \quad \forall s,\ u\in[U+1] \quad \forall t$$
$$k^t_{m,s,u}\in\{0,1\} \quad \forall m \quad \forall s,\ u\in[U] \quad \forall t$$
$$v^t_u = \|\mathbf{v}^t_u\|_2 \quad \forall u\in[U] \quad \forall t. \tag{2k}$$

1) Problem Analysis: Problem (**P2**) is a long-term stochastic MINLP with coupled continuous trajectory variables and binary offloading/relay decisions. The UAV trajectory affects link reliability, transmission rates, propulsion energy, and offloading feasibility, while offloading decisions are further constrained by cache availability and computing resources. In addition, the long-term energy budget in (2j) couples decisions across slots, making current actions affect future feasibility.

Algorithm 1 ETTJO

Input: M, U, S, T, capacity C_u, threshold ζ, EMA factor δ, learning parameters.
Output: Cache $\mathcal{B}$, offloading $\mathcal{O}$, relay decisions $\mathcal{K}$, trajectory $\mathcal{G}$.
Init: $Z_u(0)=0$, $\Psi^0=\emptyset$, $H_{\text{exp}}=0$, $\hat{H}^0_{\text{avg}}=0$, $\mathbf{s}^0_u$.
for $t=1$ *to* T **do**
  1. Event-triggered caching: Update popularity $\hat{p}^t_{m,s}$ using current requests Ω^t_m and compute H^t_{exp};
  if $t=1$ ***or*** $|H^t_{\text{exp}} - \hat{H}^{t-1}_{\text{avg}}| > \zeta$ **then**
    Execute **LSGCA (Alg. 2)** to obtain Ψ^t;
    Update $b^t_{u,s}$; set $H^t_{\text{exp}} \leftarrow F(\Psi^t)$;
  else
    $\Psi^t \leftarrow \Psi^{t-1}$ and $b^t_{u,s} \leftarrow b^{t-1}_{u,s}$;
  2. Trajectory control: for $u\in[U]$ **do**
    Observe $\mathbf{s}^t_u$; select $\mathbf{v}^t_u \leftarrow \pi_{\theta_u}(\mathbf{s}^t_u)$ via **HASAC (Alg. 4)**;
  3. Embedded offloading: Given $\mathcal{B}^t$ and $\mathbf{q}^{t+1}_u$, execute **VPDA (Alg. 3)** to obtain $\mathcal{O}^t$ and $\mathcal{K}^t$;
  Execute tasks and calculate L^t_{sys}, E^t_u, cache-hit ratio H^t, and reward r^t;
  4. Update: Update $\hat{H}^t_{\text{avg}} = \delta\hat{H}^{t-1}_{\text{avg}} + (1-\delta)H^t$ and queues according to (24);
  if *Training phase* **then**
    Store transition in $\mathcal{D}$; train **HASAC (Alg. 4)**;

Since future task arrivals, MIoT mobility, and channel states are unavailable in stochastic maritime environments, directly solving (**P2**) over the entire horizon is impractical. This motivates the online decomposition framework developed in Section V, where we introduce Lyapunov optimization to transform the energy constraint into penalties.

V. Algorithm

A. Overall Framework

To address the computational complexity and environmental stochasticity, we propose the ETTJO framework (Algorithm 1). ETTJO provides an online solution to (**P2**) in three steps. First, Lyapunov optimization transforms the long-term energy budget in (2j) into virtual energy queues, which act as dynamic energy penalties in the per-slot objective. A larger queue penalizes energy-consuming actions more strongly, thereby balancing latency reduction and energy regulation without future system information. Second, ETTJO decomposes the resulting problem into event-triggered caching, VPDA-based task offloading, and HASAC-based trajectory control. HASAC is introduced because UAV trajectory control involves continuous velocity actions, heterogeneous UAV states, and stochastic maritime mobility and channels; its entropy-regularized learning improves exploration and stabilizes policy optimization in such dynamic environments. Meanwhile, VPDA is dynamically embedded into the trajectory-learning environment: after HASAC outputs velocity actions, the induced topology is passed to VPDA, which returns feasible offloading and relay decisions. Finally, the resulting latency-energy cost is used to update the virtual queues and train the trajectory policy.

Authorized licensed use limited to: Indian Institute of Petroleum and Energy. Downloaded on October 26,2026 at 12:03:34 UTC from IEEE Xplore. Restrictions apply.

1) Lyapunov-Guided Energy Management: Lyapunov optimization is suitable for stochastic online control because it handles long-term constraints through queue stability using only current observations. For each UAV u, we introduce a virtual energy deficit queue

$$Z_u(t+1) = \max\left\{Z_u(t) + E_u^t - E_u^{\text{ref}}, 0\right\} \tag{24}$$

where $E_u^{\text{ref}} = \xi E_u^{\max}/T$ is the per-slot reference budget and $\xi \in (0,1]$ is a safety factor. A larger $Z_u(t)$ indicates a higher accumulated energy deficit and imposes a stronger penalty on future energy-consuming actions. By minimizing the Lyapunov drift-plus-penalty upper bound and removing constants independent of current decisions, (**P2**) is converted into

$$\min_{\mathcal{G}^t,\mathcal{O}^t,\mathcal{K}^t} \quad VL_{\text{sys}}^t + \sum_{u=1}^{U} Z_u(t)E_u^t \tag{P2-Online}$$

$$\text{s.t. (24a)}-\text{(24i)}, \text{(24k)}. \tag{25}$$

Here, $V \geq 0$ controls the latency–energy trade-off: a larger V emphasizes latency reduction, whereas a larger $Z_u(t)$ discourages energy-intensive decisions.

2) Two-Timescale Decomposition: Although (P2-Online) removes the temporal coupling of (2j), it remains a nonconvex mixed-integer problem involving continuous trajectory control, binary offloading/relay decisions, and cache-dependent feasibility. ETTJO therefore decomposes it into event-triggered caching and per-slot trajectory–offloading optimization.

3) Long-Term Scale (Event-Triggered Caching): The cache state is refreshed by LSGCA only when the expected cache utility deviates from the moving-average hit ratio

$$\left|F(\Psi^t) - \hat{H}_{avg}^t\right| > \zeta. \tag{26}$$

Otherwise, the previous cache state is retained to reduce unnecessary replacement and backhaul overhead.

4) Short-Term Scale (Trajectory, Offloading, and Feedback): Given the current cache state and virtual queues, HASAC selects continuous UAV velocity actions, while VPDA computes feasible discrete offloading and relay decisions under the induced topology. The realized latency, energy consumption, cache-hit observation, and reward are then used to update the virtual queues and train HASAC. This feedback links trajectory control with offloading performance, since a trajectory action can be evaluated only after VPDA determines the corresponding service paths. HASAC fits this setting because UAV velocity is continuous, UAVs may have heterogeneous observations and energy queues, and maximum-entropy learning encourages exploration under stochastic maritime dynamics.

B. Cache Update Algorithm

1) Service Popularity Estimation: Let $p_{m,s}$ denote the probability that MIoT m requests service s. Since $p_{m,s}$ is unknown, we estimate it online using the request indicator $i_{m,s}^t$

$$\widehat{p}_{m,s}^1 = i_{m,s}^1, \ \widehat{p}_{m,s}^t = \widehat{p}_{m,s}^{t-1} + \frac{1}{t}\left(i_{m,s}^t - \widehat{p}_{m,s}^{t-1}\right). \tag{27}$$

2) Submodular Utility Function: To optimize cache placement, we define the expected cache-hit utility as

$$F(\Psi^t) = \frac{\sum_{m,s} \widehat{p}_{m,s}^t \left(1 - \prod_{u\in\Psi(s)} \left(1 - l_{m,u}^t\right)\right)}{\sum_{m,s} \widehat{p}_{m,s}^t} \tag{28}$$

where $\Psi^t = \{(u,s) \mid b_{u,s}^t = 1\}$ is the set of cached UAV-service pairs.

Lemma 1: The set function $F : 2^{\mathcal{U}\times\mathcal{S}} \to \mathbb{R}_{\geq 0}$ is nondecreasing and submodular.

Proof: Please refer to the Appendix (See Supplementary Material). ∎

The marginal gain of adding a candidate pair (u,s) is

$$(u,s)_{\text{gain}} = \frac{F(\Psi \cup \{(u,s)\}) - F(\Psi)}{c_s}, \quad (u,s) \notin \Psi. \tag{29}$$

The marginal loss of removing a cached pair is

$$(u,s)_{\text{loss}} = \frac{F(\Psi) - F(\Psi \setminus \{(u,s)\})}{c_s}, \quad (u,s) \in \Psi. \tag{30}$$

3) Cache-Hit Observation and Event Trigger: The instantaneous cache-hit ratio is determined by offloading decisions. Let $\Omega_m^t = \{(m,s) \mid i_{m,s}^t = 1\}$ denote the requests generated by MIoT m at slot t. A request is counted as a cache hit if it is served by a UAV rather than relayed to the GS. Thus

$$\Omega_{m,h}^t = \left\{(m,s) \in \Omega_m^t \,\middle|\, \sum_{u=1}^{U} o_{m,s,u}^t = 1\right\} \tag{31}$$

and

$$H^t = \frac{\sum_{m\in\mathcal{M}} |\Omega_{m,h}^t|}{\sum_{m\in\mathcal{M}} |\Omega_m^t|}. \tag{32}$$

The moving-average cache-hit ratio is updated as

$$\hat{H}_{avg}^t = \delta \hat{H}_{avg}^{t-1} + (1-\delta)H^t \tag{33}$$

where $\delta \in (0,1)$ is the smoothing factor. A cache update is triggered if

$$|F(\Psi^t) - \hat{H}_{avg}^t| > \zeta. \tag{34}$$

To solve (**P1**), we propose a LSGCA, as summarized in Algorithm 2. Starting from the cache placement inherited from the previous slot, LSGCA iteratively selects uncached UAV–service pairs according to their marginal utility density. If the selected UAV has sufficient residual cache capacity, the corresponding service is directly cached. Otherwise, LSGCA constructs a tentative replacement by evicting cached services with the smallest marginal loss density from the same UAV until enough cache space is released. The replacement is accepted only when the marginal utility gain of the new service exceeds the total utility loss caused by eviction. In this way, LSGCA enables adaptive cache updates under limited storage while avoiding unnecessary replacement of valuable cached services.

Theorem 1: The LSGCA 2 achieves an approximation ratio of at least 1/2. That is, if Ψ is the set returned by Algorithm 2 and Ψ^* is the optimal set maximizing F under the knapsack constraint, then

$$F(\Psi) \geq \frac{1}{2}F(\Psi^*). \tag{35}$$

Proof: Please refer to the Appendix (See Supplementary Material). ∎

Authorized licensed use limited to: Indian Institute of Petroleum and Energy. Downloaded on October 26,2026 at 12:03:34 UTC from IEEE Xplore. Restrictions apply.

Algorithm 2 LSGCA

Input: previous cache state $\{b_{u,s}^{t-1}\}$, popularity $\{\widehat{p}_{m,s}^{t}\}$, and link reliabilities $\{l_{m,u}^{t}\}$.
Output: Caching decisions $b_{u,s}^{t} \in \{0,1\}$.
Init: $\Psi \leftarrow \{(u,s) \mid b_{u,s}^{t-1} = 1\}$, $b_{u,s}^{t} \leftarrow 0$;
$\mathcal{V} \leftarrow \{(u,s) \mid u \in [U], s \in [S]\} \setminus \Psi$, $\forall u, s$;
$W_u \leftarrow C_u - \sum_{s:(u,s)\in\Psi} c_s$, $\forall u$;

while $|\mathcal{V}| > 0$ **do**
  $(\hat{u}, \hat{s}) \leftarrow \arg\max_{(u,s)\in\mathcal{V}} \frac{F(\Psi\cup\{(u,s)\})-F(\Psi)}{c_s}$;
  $gain \leftarrow F(\Psi \cup \{(\hat{u},\hat{s})\}) - F(\Psi)$;
  if $W_{\hat{u}} \geq c_{\hat{s}}$ **then**
    $W_{\hat{u}} \leftarrow W_{\hat{u}} - c_{\hat{s}}$, $\Psi \leftarrow \Psi \cup \{(\hat{u},\hat{s})\}$;
  else
    $\Psi' \leftarrow \Psi$; $W' \leftarrow W_{\hat{u}}$; $loss \leftarrow 0$;
    while $W' < c_{\hat{s}}$ **and** $gain > loss$ **do**
      $(\hat{u}, \dot{s}) \leftarrow \arg\min_{(\hat{u},s)\in\Psi'} \frac{F(\Psi')-F(\Psi'\setminus\{(\hat{u},s)\})}{c_s}$;
      $loss \leftarrow loss + (F(\Psi') - F(\Psi' \setminus \{(\hat{u},\dot{s})\}))$;
      $\Psi' \leftarrow \Psi' \setminus \{(\hat{u},\dot{s})\}$, $W' \leftarrow W' + c_{\dot{s}}$;
    if $W' \geq c_{\hat{s}}$ **and** $gain > loss$ **then**
      $\Psi \leftarrow \Psi' \cup \{(\hat{u},\hat{s})\}$, $W_{\hat{u}} \leftarrow W' - c_{\hat{s}}$;
  $\mathcal{V} \leftarrow \mathcal{V} \setminus \{(\hat{u},\hat{s})\}$;
for $(u,s) \in \Psi$ **do** $b_{u,s}^{t} \leftarrow 1$;

Algorithm 3 VPDA

$\mathcal{T}_{\text{unmatch}} \leftarrow \mathcal{T}$; $\mathcal{M}_u \leftarrow \emptyset, \forall u$; $\mathcal{D}_{\text{drop}}^{t} \leftarrow \emptyset$; $\mathbf{o}^t, \mathbf{k}^t \leftarrow 0$.
Compute costs $\Omega_{m,s}(p)$ and preference lists $\text{PL}_{(m,s)}$, $\forall (m,s)$;
while $\mathcal{T}_{\text{unmatch}} \neq \emptyset$ **do**
  Pop (m,s) from $\mathcal{T}_{\text{unmatch}}$;
  if $\text{PL}_{(m,s)} = \emptyset$ **then**
    $\mathcal{D}_{\text{drop}}^{t} \leftarrow \mathcal{D}_{\text{drop}}^{t} \cup \{(m,s)\}$; **continue**;
  Select best $p^* \in \text{PL}_{(m,s)}$ and remove it from $\text{PL}_{(m,s)}$;
  Map p^* to UAV u and set $\mathcal{S} \leftarrow \mathcal{M}_u \cup \{(m,s,p^*)\}$;
  if $\mathcal{S}$ *violates constraints* (2d)–(2i) **then**
    $(m',s',p') \leftarrow \arg\max_{(i,j,\hat{p})\in\mathcal{S}} \Omega_{i,j}(\hat{p})$;
    $\mathcal{M}_u \leftarrow \mathcal{S} \setminus \{(m',s',p')\}$;
    $\mathcal{T}_{\text{unmatch}} \leftarrow \mathcal{T}_{\text{unmatch}} \cup \{(m',s')\}$;
  else
    $\mathcal{M}_u \leftarrow \mathcal{S}$;
for *each* u *and* $(m,s,p) \in \mathcal{M}_u$ **do**
  if $p = u_{\text{cmp}}$ **then**
    $o_{m,s,u}^{t} \leftarrow 1$;
  else if $p = u_{\text{rel}}$ **then**
    $k_{m,s,u}^{t} \leftarrow 1$; $o_{m,s,U+1}^{t} \leftarrow 1$;
return $\mathbf{o}^t$, $\mathbf{k}^t$, $\mathcal{D}_{\text{drop}}^{t}$, *and the latency-energy cost with drop penalties.*

C. Task-Offloading Decision

1) Task Offloading Subproblem: Based on (P2-Online), we first derive the discrete task-offloading subproblem solved by VPDA. Given fixed cache placement and UAV trajectories, link reliabilities, transmission rates, and cache placement become known. The terms independent of offloading and relay decisions can then be omitted, yielding the following integer linear programming (ILP).

Lemma 2: Given fixed trajectory and caching states, the task offloading subproblem derived from (P2-Online) is equivalent to an ILP problem

$$\min_{\boldsymbol{o}^t,\boldsymbol{k}^t} \quad \sum_{m=1}^{M}\sum_{s=1}^{S}\sum_{u=1}^{U}\left(Q_{m,s,u}^{t} o_{m,s,u}^{t} + R_{m,s,u}^{t} k_{m,s,u}^{t}\right) \tag{P2.1}$$

s.t. Constraints (24d)–(24i)

$$o_{m,s,u}^{t}, k_{m,s,u}^{t} \in \{0,1\} \quad \forall m \quad \forall s \quad \forall u \in [U] \quad \forall t \tag{2.1a}$$

where $Q_{m,s,u}^{t}$ and $R_{m,s,u}^{t}$ are deterministic cost coefficients for direct UAV computing and UAV-to-GS relaying, respectively.

Proof: Please refer to the Appendix (See Supplementary Material). ∎

2) Matching Game Formulation [35]: We recast the ILP problem (P2.1) as a many-to-one matching game with externalities, defined by the tuple $(\mathcal{T}, \mathcal{P}, \succ)$, where $\mathcal{T}$ denotes the task set and $\mathcal{P}$ denotes the virtual service-port set. Each UAV u is abstracted into a computation port u_{cmp} and a relay port u_{rel}, giving

$$\mathcal{P} = \{1_{\text{cmp}}, \ldots, U_{\text{cmp}}\} \cup \{1_{\text{rel}}, \ldots, U_{\text{rel}}\}.$$

Each task (m,s) constructs a preference list $\mathcal{PL}_{(m,s)}$ over $\mathcal{P}$. The task-specific cost mapping is

$$\Omega_{m,s}(p) = \begin{cases} Q_{m,s,u}^{t}, & p = u_{\text{cmp}} \\ R_{m,s,u}^{t}, & p = u_{\text{rel}}. \end{cases} \tag{36}$$

Thus,

$$p_i \succ_{(m,s)} p_j \iff \Omega_{m,s}(p_i) < \Omega_{m,s}(p_j). \tag{37}$$

3) VPDA: Tasks propose to virtual ports according to their preference lists. Physical UAVs act as arbiters and reject tentative proposals that violate cache availability, link reliability, relay feasibility, or computation-capacity constraints in (2d)–(2i).

Theorem 2: The VPDA algorithm converges to a pairwise stable matching $\mathcal{M}$, constituting a Nash Equilibrium where no task can unilaterally reduce its cost without violating system constraints.

Proof: Please refer to the Appendix (See Supplementary Material). ∎

D. Trajectory Plan

1) Trajectory Plan Formulation: Trajectory planning is coupled with task offloading because UAV positions determine link reliability, transmission rates, and feasible service paths. At each slot, HASAC selects UAV velocity actions, which induce the post-action topology at $\mathbf{q}_u^{t+1}$. Given the current cache state and updated topology, VPDA returns the corresponding offloading and relay decisions, i.e., $o_{m,s,u}^{t}$ and $k_{m,s,u}^{t}$. Substituting these VPDA decisions into (P2-Online), the analytical trajectory subproblem retains trajectory-dependent communication and flight costs, while computation-related terms are omitted because they are fixed under a given VPDA assignment. During HASAC training, VPDA is re-run after each sampled trajectory action, and the realized feedback includes communication, computation, relay, and drop-penalty costs. The per-slot trajectory subproblem is formulated as

$$\min_{\mathcal{G}^t} \quad \sum_{u=1}^{U}\left(\Phi_{u,\text{tx}}^{t}(\mathbf{q}_u^{t+1}) + \Phi_{u,\text{fly}}^{t}(\mathbf{v}_u^{t})\right) \tag{P2.2}$$

Authorized licensed use limited to: Indian Institute of Petroleum and Energy. Downloaded on October 26,2026 at 12:03:34 UTC from IEEE Xplore. Restrictions apply.

$$\text{s.t. } \|\mathbf{v}_u^t\|_2 \le V_u \quad \forall u \in [U], \tag{2.2a}$$

$$\left\|\mathbf{q}_j^{t+1} - \mathbf{q}_k^{t+1}\right\|_2 \ge d_{\min} \quad \forall j,k \in [U], j \ne k \tag{2.2b}$$

$$\prod_{u=1}^{U}(1 - l_{m,u}^{t+1}) \le \epsilon \quad \forall m \tag{2.2c}$$

$$\mathbf{q}_u^{t+1} = \mathbf{q}_u^t + \mathbf{v}_u^t \Delta t \quad \forall u \in [U] \tag{2.2d}$$

where $\mathcal{G}^t = \{\mathbf{v}_u^t \mid u \in [U]\}$ denotes the UAV velocity decisions. The superscript $t+1$ denotes the post-action topology within slot t, while requests, cache states, VPDA decisions, and virtual queues remain indexed by t.

The communication cost is

$$\Phi_{u,\text{tx}}^t\left(\mathbf{q}_u^{t+1}\right) = \sum_{m=1}^{M}\sum_{s=1}^{S} V n_{m,s}^t \left(\frac{o_{m,s,u}^t + k_{m,s,u}^t}{R_{m,u}^{t+1}} + \frac{k_{m,s,u}^t}{R_{u,g}^{t+1}}\right) + \sum_{m=1}^{M}\sum_{s=1}^{S} Z_u(t) P_{\text{tx},u}^t k_{m,s,u}^t \frac{n_{m,s}^t}{R_{u,g}^{t+1}}. \tag{38}$$

The flight cost is

$$\Phi_{u,\text{fly}}^t(\mathbf{v}_u^t) = Z_u(t) P_{\text{fly},u}\left(v_u^t\right) \Delta t. \tag{39}$$

Therefore, UAV velocity actions affect the updated topology, VPDA assignment, relay selection, and the resulting latency-energy cost.

2) Markov Decision Process: Since (**P2.2**) involves continuous velocity control and embedded VPDA responses, it is difficult to solve analytically. We formulate trajectory planning as a multiagent MDP, where HASAC outputs continuous UAV velocity actions, the environment invokes VPDA under the updated topology, and the resulting latency-energy cost is returned as reward feedback during training. The MDP is defined by $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{D}, \mathcal{R}, \gamma \rangle$.

3) Observation Space ($\mathcal{S}$): The local observation $\mathbf{s}_u^t$ for UAV u at time slot t captures the agent's kinematic state and its relative geometry to the environment

$$\mathbf{s}_u^t = \left\{\mathbf{q}_u^t, \mathbf{v}_u^t, Z_u(t), \quad \left\{\mathbf{d}_{u,k}^t\right\}_{k\in\mathcal{K}_u}, \quad \left\{\mathbf{v}_m^t, \mathcal{F}_m^t\right\}_{m\in[M]}\right\} \tag{40}$$

where $\mathcal{K}_u = \{g\} \cup [M] \cup ([U] \setminus \{u\})$, $\mathbf{d}_{u,k}^t$ denotes the relative position from UAV u to node k, and $\mathcal{F}_m^t$ denotes the current task-request feature of MIoT device m.

4) Action Space ($\mathcal{A}$): Each UAV chooses a continuous velocity action sampled from the actor network $\mathbf{a}_u^t \sim \pi_{\theta_u}(\cdot \mid \mathbf{s}_u^t)$

$$a_u^t = \mathbf{v}_u^t, \quad \mathcal{A}_u = \{\mathbf{v} \mid \|\mathbf{v}\|_2 \le V_u\}. \tag{41}$$

5) Transition Dynamics ($\mathcal{D}$): The state evolves according to UAV movement, MIoT mobility, VPDA-based offloading feedback, and virtual queue updates. In particular, UAV positions follow (2.2d), while energy queues follow (24).

6) Reward Function ($\mathcal{R}$): The global reward is defined as

$$r_{\text{total}}^t = -\sum_{u=1}^{U}\left(\Phi_{u,\text{tx}}^t\left(\mathbf{q}_u^t\right) + \Phi_{u,\text{fly}}^t\left(v_u^t\right)\right) - \lambda_1 C_{\text{safe}}^t - \lambda_2 C_{\text{link}}^t \tag{42}$$

where $\lambda_1, \lambda_2 > 0$ are penalty coefficients. The penalties are

$$C_{\text{safe}}^t = \sum_{j=1}^{U}\sum_{k=j+1}^{U} \mathbb{I}\left(\left\|\mathbf{q}_j^{t+1} - \mathbf{q}_k^{t+1}\right\|_2 < d_{\min}\right) \tag{43}$$

Algorithm 4 Heterogeneous-Agent Soft Actor-Critic (HASAC)

Input: Temperature α, Polyak coefficient τ, batch size B, agents U, episodes K, horizon T.
Init: Critics ϕ_1, ϕ_2, policies $\{\theta_u\}_{u\in[U]}$, replay buffer $\mathcal{D}$, and target critics $\phi_{\text{targ},j} \leftarrow \phi_j$, $j = 1, 2$.

```
for k = 1 to K do
    for t = 1 to T do
        Each agent u observes s_u^t and samples a_u^t ~ π_{θ_u}(· | s_u^t)
        Execute a^t, obtain rewards and next states, and store the transition in D
    for each gradient update step do
        Sample a minibatch from D and compute target y using (46)
        Update critics φ_j, j = 1, 2, by minimizing L(φ_j) = (1/B) Σ (y − Q_{φ_j}(s, a))^2
        Draw a random permutation P and update each policy θ_n, n ∈ P, by minimizing (47)
        Update target critics: φ_{targ,j} ← τφ_j + (1 − τ)φ_{targ,j}, j = 1, 2
```

$$C_{\text{link}}^t = \sum_{m} \mathbb{I}\left(\prod_{u=1}^{U}\left(1 - l_{m,u}^{t+1}\right) > \epsilon\right). \tag{44}$$

7) Learning Objective: HASAC maximizes the entropy-regularized return

$$J(\boldsymbol{\pi}) = \mathbb{E}_{\mathbf{a}\sim\boldsymbol{\pi}}\left[\sum_{t=0}^{T}\gamma^t\left(r^t + \alpha\sum_{u=1}^{U}\mathcal{H}\left(\pi_u\left(\cdot \mid \mathbf{s}_u^t\right)\right)\right)\right] \tag{45}$$

where γ is the discount factor, α is the temperature parameter, and $\mathcal{H}(\cdot)$ denotes policy entropy.

8) Centralized Soft-Q Function: Under CTDE, the target soft-Q value is

$$y^t = r^t + \gamma\left(\min_{j=1,2} Q_{\phi_{\text{targ},j}}\left(\mathbf{s}^{t+1}, \mathbf{a}^{t+1}\right) - \alpha\sum_{u=1}^{U}\log\pi_{\theta_u}\left(\mathbf{a}_u^{t+1} \mid \mathbf{s}_u^{t+1}\right)\right). \tag{46}$$

9) Decentralized Policy Function: HASAC updates agents sequentially. For agent n, the policy loss is

$$J(\theta_n) = \mathbb{E}\left[\alpha\log\pi_{\theta_n}\left(\mathbf{a}_n^t \mid \mathbf{s}_n^t\right) - \min_{j=1,2} Q_{\phi_j}\left(\mathbf{s}^t, \mathbf{a}_{1:n-1}^{\text{new}}, \mathbf{a}_n^t, \mathbf{a}_{n+1:U}^{\text{old}}\right)\right]. \tag{47}$$

Theorem 3 (Heterogeneous-Agent Soft Policy Iteration [36]): Under the assumptions in [36], repeated joint soft policy evaluation and heterogeneous-agent soft policy improvement lead the joint policy $\boldsymbol{\pi} = \prod_{u=1}^{U}\pi_u$ to converge to a Quantal Response Equilibrium.

VI. Performance Evaluation

A. Simulation Setup

We evaluate ETTJO in an offshore maritime MEC scenario, where UAVs provide cache-assisted edge computing for heterogeneous MIoT devices, including vessels and buoys. Device

Authorized licensed use limited to: Indian Institute of Petroleum and Energy. Downloaded on October 26,2026 at 12:03:34 UTC from IEEE Xplore. Restrictions apply.

TABLE I
SIMULATION PARAMETERS

Parameter	Value
System & Channel	
Area Size / Horizon (T)	4×4 km^2 / 200 slots
Slot Duration (Δt)	10 s
Bandwidth / Noise	1 MHz / -114 dBm
Reliability / Zipf	$(\epsilon, \chi) = (0.1, 0.5)$ / 0.65
UAV (DJI Mavic 4 Pro)	
Max Speed / Battery	50 m/s / 95 Wh
Coverage Radius / Altitude	1000 m / 100 m
Flying Coef. (C_1, C_2, C_3)	70, 0.021, 57
Flying Coef. (C_4, C_5)	38, 0.015
User & Task	
Users (M) / UAVs (U)	[12, 42] / [4, 12]
Services (S) / Cache (C_u)	[4, 20] / [2, 6]
Velocity (Vessel/Buoy)	5 / 0.5 m/s
Tx Power (Vessel/Buoy)	10 / 0.01 W
Task Size	[100, 1000] KB

trajectories follow linear mobility with boundary turnover, and service popularity follows a Zipf distribution [22]. Real AIS traces are left for future work. The probabilistic LoS channel model in [33] is adopted to capture maritime air-to-ground propagation. Unless otherwise specified, the default setting is $C_u = 4$, $S = 8$, $U = 6$, and $M = 24$, denoting the UAV cache capacity, service types, UAVs, and MIoT devices, respectively. The simulation horizon contains $T = 200$ slots with $\Delta t = 10$ s. UAV propulsion energy is calculated using DJI Mavic 4 Pro specifications [37]. All schemes use identical traces, and learning-based results are averaged over five independent runs with the same seed set. The control and learning parameters are set as $V = 10^2$, $\xi = 0.9$, $\zeta = 0.05$, $\delta = 0.9$, lr $= 10^{-4}$, and batch size 512. The main simulation parameters are summarized in Table I.

To evaluate ETTJO, we compare it with four representative schemes.

1) RANDOM is a lower-bound benchmark where caching and trajectories are randomized, and offloading follows a greedy feasible assignment.
2) Greedy-based is a low-complexity heuristic where UAVs cache historically popular services, move toward demand hotspots, and offload tasks to the lowest-latency node.
3) TJSO [6] is a literature-based joint trajectory and service offloading scheme adapted to the considered maritime UAV-MEC scenario, with periodic cache updates.
4) MAPPO [38] is a representative multiagent reinforcement learning baseline using centralized training and decentralized execution for UAV coordination. For clarity, ETTJO is labeled as "Ours" in the figures.

B. Evaluation Results

1) Main Performance: Figs. 2–4 compare the delay, cumulative energy consumption, and cache hit ratio of different schemes under the default setting. RANDOM suffers from the largest delay because random caching and trajectory decisions cannot maintain stable service coverage. Greedy-based achieves low delay in early slots by aggressively moving toward service-demand hotspots and selecting instantaneous minimum-latency service paths. However, its delay increases sharply in later slots due to excessive energy consumption and degraded service capability. TJSO and MAPPO provide more stable performance than Greedy-based, but they still

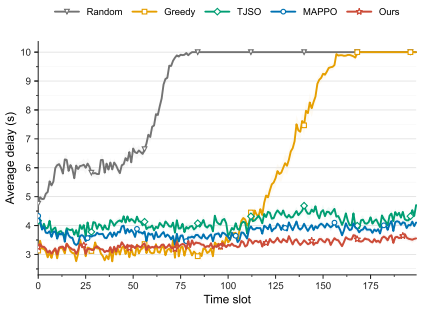


Fig. 2. Average delay comparison under the default setting.

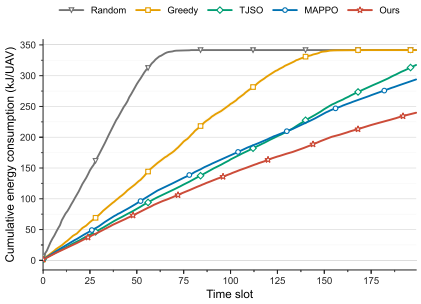


Fig. 3. Cumulative energy consumption under the default setting.

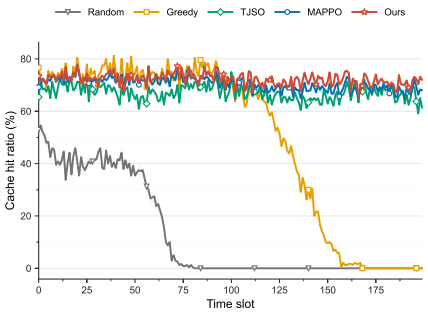


Fig. 4. Cache hit ratio under the default setting.

cannot fully exploit the proposed event-triggered caching and Lyapunov-guided online control. In contrast, ETTJO maintains low and stable delay over the whole horizon while consuming energy more gradually.

Fig. 3 shows that ETTJO consumes less cumulative energy than TJSO and MAPPO while maintaining lower delay. This indicates that ETTJO does not reduce delay by excessive UAV movement or energy-consuming service decisions. Fig. 4 further shows that ETTJO achieves a high cache hit ratio in most slots, verifying the effectiveness of the event-triggered caching strategy.

2) Impact of Cache Capacity and Service Diversity: Fig. 5 shows the impact of UAV cache capacity. As C_u increases, all schemes achieve lower delay because more services can be stored at UAVs and more tasks can be served locally. ETTJO achieves the best overall performance, indicating that the proposed event-triggered caching and offloading coordination can better exploit increased cache resources.

Fig. 6 shows the impact of the number of service types. The delay increases with S because a larger service set intensifies cache competition under limited UAV storage. The performance gap between ETTJO and other schemes becomes more

Authorized licensed use limited to: Indian Institute of Petroleum and Energy. Downloaded on October 26,2026 at 12:03:34 UTC from IEEE Xplore. Restrictions apply.

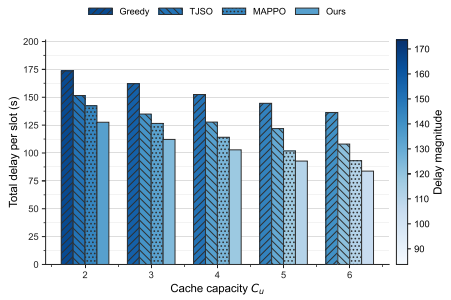


Fig. 5. Total delay per slot versus cache capacity C_u.

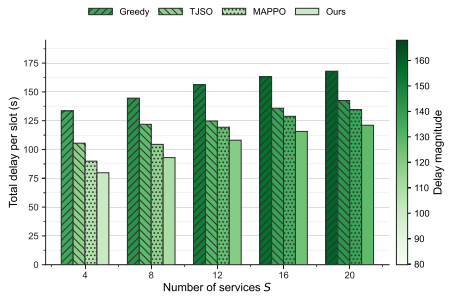


Fig. 6. Total delay per slot versus the number of service types S.

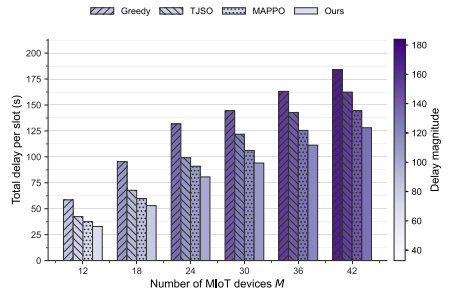


Fig. 7. Total delay per slot versus the number of MIoT devices M.

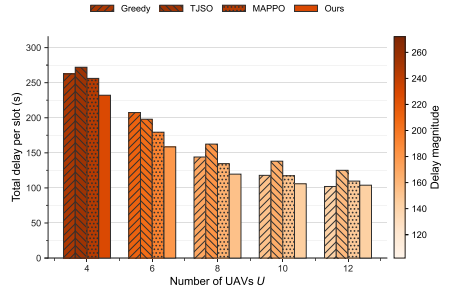


Fig. 8. Total delay per slot versus the number of UAVs U.

evident as S increases, demonstrating the robustness of the proposed caching mechanism under service-diverse maritime task demands. RANDOM is omitted from this sensitivity comparison for readability because it mainly serves as a lower-bound reference in Fig. 2.

3) Scalability: Fig. 7 evaluates the impact of the number of MIoT devices. As M increases, the system delay increases for all schemes because more tasks compete for limited UAV computation, caching, and communication resources. ETTJO achieves the lowest delay in the tested MIoT range, showing its scalability under heavier maritime traffic loads.

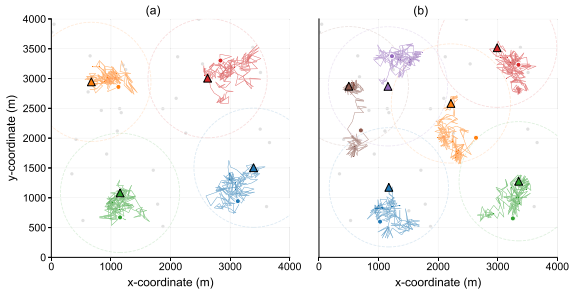


Fig. 9. Trajectory visualization of ETTJO under representative UAV scales. (a) $U = 4$. (b) $U = 6$.

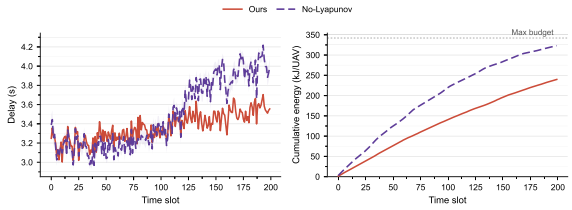


Fig. 10. Ablation comparison between ETTJO and the No-Lyapunov variant in terms of delay and energy consumption.

Fig. 8 shows the impact of the number of UAVs. The delay generally decreases as U increases, since more UAVs provide wider coverage, shorter access distances, and more edge-computing resources. As U further increases, the improvement becomes less significant, showing diminishing returns from additional UAV deployment. ETTJO achieves the lowest delay in most cases and outperforms TJSO and MAPPO. When $U = 12$, Greedy-based is slightly better because each UAV serves only about two MIoT devices on average, and thus moving directly toward current demand areas is already effective. By contrast, ETTJO also considers energy regulation and coverage distribution, which may lead to a slightly higher delay in this resource-sufficient case. RANDOM is omitted for readability.

4) Trajectory Visualization: Fig. 9 visualizes the UAV trajectories generated by ETTJO under two representative UAV scales, i.e., $U = 4$ and $U = 6$. The UAVs adaptively move toward request-intensive maritime areas while maintaining distributed coverage over the service region. Compared with the case of $U = 4$, increasing the number of UAVs to $U = 6$ enables a finer spatial division of the service area, allowing UAVs to cover geographically dispersed MIoT devices with less redundant movement. This observation is consistent with the delay reduction reported in the scalability analysis, since more UAVs can shorten access distances and improve service availability.

5) Effect of Lyapunov-Guided Energy Regulation: Fig. 10 compares ETTJO with the No-Lyapunov variant in terms of delay and cumulative energy consumption. No-Lyapunov removes the virtual energy queues while keeping other ETTJO modules unchanged. This ablation evaluates the contribution of Lyapunov-guided energy regulation. Without virtual energy queues, No-Lyapunov can still implicitly learn certain energy-saving behaviors through long-term rewards, but such delayed feedback leads to larger latency fluctuations and faster energy consumption. By using virtual energy queues as explicit online penalty weights, ETTJO regulates energy-consuming actions and achieves a more stable latency-energy trade-off.

Authorized licensed use limited to: Indian Institute of Petroleum and Energy. Downloaded on October 26,2026 at 12:03:34 UTC from IEEE Xplore. Restrictions apply.

VII. Conclusion

This article proposed ETTJO, an event-triggered two-timescale framework for joint caching, trajectory control, and task offloading in UAV-assisted maritime MEC. By combining Lyapunov-guided energy regulation, event-triggered cache updates, VPDA-based task offloading, and HASAC-based trajectory control, ETTJO addresses stochastic maritime dynamics while balancing latency and UAV energy consumption. Theoretical analysis establishes key properties of the proposed caching, offloading, and learning components. Simulation results show that ETTJO reduces service delay in most tested scenarios, improves cache-hit performance, and achieves more sustainable energy consumption compared with representative baselines.

References

[1] F. S. Alqurashi, A. Trichili, N. Saeed, B. S. Ooi, and M.-S. Alouini, "Maritime communications: A survey on enabling technologies, opportunities, and challenges," *IEEE Internet Things J.*, vol. 10, no. 4, pp. 3525–3547, Feb. 2023.

[2] S. Rauniyar, P. Orten, and S. Petersen, "Improving last-mile maritime communication using intelligent reflecting surfaces," in *Proc. 34th Int. Telecommun. Netw. Appl. Conf. (ITNAC)*, Nov. 2024, pp. 1–6.

[3] Z. Wang, B. Lin, Q. Ye, Y. Fang, and X. Han, "Joint computation offloading and resource allocation for maritime MEC with energy harvesting," *IEEE Internet Things J.*, vol. 11, no. 11, pp. 19898–19913, Jun. 2024.

[4] M. Yan, H. Guo, C. A. Chan, A. F. Gygax, C. Li, and I. Chih-Li, "Semantic communication-enabled multi-access edge computing network resource optimization in the 6G era," *IEEE Wireless Commun.*, vol. 33, no. 2, pp. 83–91, Apr. 2026.

[5] T. Wei, W. Feng, J. Wang, N. Ge, and J. Lu, "Exploiting the shipping lane information for energy-efficient maritime communications," *IEEE Trans. Veh. Technol.*, vol. 68, no. 7, pp. 7204–7208, Jul. 2019.

[6] R. Zhou, X. Wu, H. Tan, and R. Zhang, "Two time-scale joint service caching and task offloading for UAV-assisted mobile edge computing," in *Proc. IEEE INFOCOM - IEEE Conf. Comput. Commun.*, May 2022, pp. 1189–1198.

[7] Y. Qu et al., "Service provisioning for UAV-enabled mobile edge computing," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 11, pp. 3287–3305, Nov. 2021.

[8] L. Wang, K. Wang, C. Pan, W. Xu, N. Aslam, and A. Nallanathan, "Deep reinforcement learning based dynamic trajectory control for UAV-assisted mobile edge computing," *IEEE Trans. Mobile Comput.*, vol. 21, no. 10, pp. 3536–3550, Oct. 2022.

[9] R. Raulefs, M. A. Bellido-Manganell, A. Dammann, M. Walter, and W. Wang, "Verification and modeling of the maritime channel for maritime communications and navigation networks," *Frontiers Mar. Sci.*, vol. 10, Jun. 2023, Art. no. 1158524.

[10] R. Zhou et al., "User preference oriented service caching and task offloading for UAV-assisted MEC networks," *IEEE Trans. Services Comput.*, vol. 18, no. 2, pp. 1097–1109, Mar. 2025.

[11] F. Lu et al., "Resource and trajectory optimization for UAV-relay-assisted secure maritime MEC," *IEEE Trans. Commun.*, vol. 72, no. 3, pp. 1641–1652, Mar. 2024.

[12] Y. Zhang, Z. Kuang, Y. Feng, and F. Hou, "Task offloading and trajectory optimization for secure communications in dynamic user multi-UAV MEC systems," *IEEE Trans. Mobile Comput.*, vol. 23, no. 12, pp. 14427–14440, Dec. 2024.

[13] Y. Wu, S. Yao, Y. Yang, Z. Hu, and C.-X. Wang, "Semigradient-based cooperative caching algorithm for mobile social networks," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Dec. 2016, pp. 1–6.

[14] A. Malik, J. Kim, K. S. Kim, and W.-Y. Shin, "A personalized preference learning framework for caching in mobile networks," *IEEE Trans. Mobile Comput.*, vol. 20, no. 6, pp. 2124–2139, Jun. 2021.

[15] S. Khuller, A. Moss, and J. S. Naor, "The budgeted maximum coverage problem," *Inf. Process. Lett.*, vol. 70, no. 1, pp. 39–45, Apr. 1999.

[16] M. Wu, H. Wu, W. Lu, L. Guo, I. Lee, and A. Jamalipour, "Security-aware designs of multi-UAV deployment, task offloading and service placement in edge computing networks," *IEEE Trans. Mobile Comput.*, vol. 24, no. 10, pp. 11046–11060, Oct. 2025.

[17] R. Zhang, R. Lu, X. Cheng, N. Wang, and L. Yang, "A UAV-enabled data dissemination protocol with proactive caching and file sharing in V2X networks," *IEEE Trans. Commun.*, vol. 69, no. 6, pp. 3930–3942, Jun. 2021.

[18] G. T. Maale, N. A. E. Kuadey, Y. Arafat, K. Thomas, G. Sun, and G. Liu, "Multi-task learning for UAV trajectory and caching with federated cloud-assisted knowledge distillation," *IEEE Trans. Netw. Service Manage.*, vol. 22, no. 3, pp. 2516–2533, Jun. 2025.

[19] J. Zhang, F. Liang, B. Li, Z. Yang, Y. Wu, and H. Zhu, "Placement optimization of caching UAV-assisted mobile relay maritime communication," *China Commun.*, vol. 17, no. 8, pp. 209–219, Aug. 2020.

[20] J. You, Z. Jia, C. Dong, Q. Wu, and Z. Han, "Joint computation offloading and resource allocation for uncertain maritime MEC via cooperation of UAVs and vessels," 2025, *arXiv:2506.15225*.

[21] M. Yan, P. Yu, J. Chen, C. Li, and I. Chih-Li, "Dynamic computation offloading optimization based on meta-reinforcement learning in UAV-assisted MEC networks," *IEEE Internet Things J.*, vol. 13, no. 11, pp. 23615–23627, Jun. 2026.

[22] B. Li et al., "UAV-enabled joint sensing, communication, powering and backhaul transmission in maritime monitoring networks," 2025, *arXiv:2505.12190*.

[23] J. Bai et al., "The DDPG-based joint optimization of task offloading and content caching in UAV-assisted IoV," *IEEE Internet Things J.*, vol. 12, no. 19, pp. 40330–40346, Oct. 2025.

[24] M. Xie, L. Li, and X. Ni, "Cache-assisted task offloading strategy based on multi-agent deep reinforcement learning," *Comput. Netw.*, vol. 270, Mar. 2025, Art. no. 111483.

[25] F. Song et al., "Multi-objective dependent task scheduling, resource allocation, and service caching in aerial-ground integrated MEC," *IEEE Trans. Intell. Transp. Syst.*, vol. 26, no. 9, pp. 13489–13505, Sep. 2025.

[26] Y. Li, H. Wang, J. Sun, H. Lv, W. Zheng, and G. Feng, "Two-timescale joint service caching and resource allocation for task offloading with edge–cloud cooperation," *Comput. Netw.*, vol. 254, Dec. 2024, Art. no. 110771.

[27] P. Qin, X. Wu, M. Fu, R. Ding, and Y. Fu, "Latency minimization resource allocation and trajectory optimization for UAV-assisted cache-computing network with energy recharging," *IEEE Trans. Commun.*, vol. 73, no. 8, pp. 5715–5728, Aug. 2025.

[28] X. Gao and L. Zhai, "Service experience oriented cooperative computing in cache-enabled UAVs assisted MEC networks," *IEEE Trans. Mobile Comput.*, vol. 23, no. 10, pp. 9721–9736, Oct. 2024.

[29] Q. Tang, S. Wen, S. He, and K. Yang, "Multi-UAV-assisted offloading for joint optimization of energy consumption and latency in mobile edge computing," *IEEE Syst. J.*, vol. 18, no. 2, pp. 1414–1425, Jun. 2024.

[30] Z. Wang, T. Wei, G. Sun, X. Liu, H. Yu, and D. Niyato, "Multi-UAV enabled MEC networks: Optimizing delay through intelligent 3D trajectory planning and resource allocation," 2024, *arXiv:2409.17882*.

[31] H. Gong, B. Huang, B. Jia, and H. Dai, "Modeling power consumptions for multirotor UAVs," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 6, pp. 7409–7422, Dec. 2023.

[32] Y. Zeng, J. Xu, and R. Zhang, "Energy minimization for wireless communication with rotary-wing UAV," *IEEE Trans. Wireless Commun.*, vol. 18, no. 4, pp. 2329–2345, Apr. 2019.

[33] Y. Wang, W. Feng, J. Wang, and T. Q. S. Quek, "Hybrid satellite-UAV-terrestrial networks for 6G ubiquitous coverage: A maritime communications perspective," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 11, pp. 3475–3490, Nov. 2021.

[34] Q. Luo, T. H. Luan, W. Shi, and P. Fan, "Deep reinforcement learning based computation offloading and trajectory planning for multi-UAV cooperative target search," *IEEE J. Sel. Areas Commun.*, vol. 41, no. 2, pp. 504–520, Feb. 2023.

[35] D. Niyato and E. Hossain, "Competitive pricing for spectrum sharing in cognitive radio networks: Dynamic game, inefficiency of Nash equilibrium, and collusion," *IEEE J. Sel. Areas Commun.*, vol. 26, no. 1, pp. 192–202, Jan. 2008.

[36] J. Liu et al., "Maximum entropy heterogeneous-agent reinforcement learning," in *Proc. 12th Int. Conf. Learn. Represent.*, 2024, pp. 1–12. [Online]. Available: https://openreview.net/forum?id=tmqOhBC4a5

[37] (2025). *DJI Mavic 4 Pro: Tri-Camera Flagship Aerial Camera Drone*. Accessed: Dec. 21, 2025. [Online]. Available: https://www.dji.com/global/mavic-4-pro

[38] C. Yu et al., "The surprising effectiveness of PPO in cooperative multi-agent games," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 24611–24624.

Authorized licensed use limited to: Indian Institute of Petroleum and Energy. Downloaded on October 26,2026 at 12:03:34 UTC from IEEE Xplore. Restrictions apply.